

Project Design Phase-II

Data Flow Diagram & User Stories

- **Date:** 18 June 2026
- **Project Name:** Shopez

Data Flow Diagrams (DFD)

Level 0 DFD (Context Diagram)

- **Entities:** Customer, Administrator.
- **Inputs:** Credentials, Cart selection, Payment details (Customer); Inventory updates (Admin).
- **Outputs:** Order confirmation, product views (Customer); Stats reports, order logs (Admin).

Level 1 DFD (Process Level)

1. **User Authentication Process:** Inputs credentials -> Queries MongoDB -> Returns JWT.
2. **Product Catalog Process:** Fetches product listings -> Loads from MongoDB (or reads fallback JSON) -> Displays to UI.
3. **Cart & Checkout Process:** Processes products -> Computes totals -> Simulates credit card verification -> Decrements stock -> Saves order.
4. **Admin Panel Process:** Calculates metrics (Sales, Orders, Stock levels) -> Renders insights.

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer	Registration	USN-1	As a user, I can register by entering email and password	Redirects to login, password is encrypted	High	Sprint-1
Customer	Checkout	USN-2	As a user, I can pay with an animated credit card UI	Credit card dynamically reflects name and card numbers	High	Sprint-2
Customer	Shopping	USN-3	As a user, I want to add products to my cart	Cart counters increment immediately	High	Sprint-1
Administrator	Dashboard	USN-4	As an admin, I can view overall	Sales charts and statistics are easily readable	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
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Administrator	Inventory	USN-5	As an admin, I can manage products (Add/Edit/Delete)	Catalog is updated in MongoDB/JSON immediately	Medium	Sprint-2